MX

In [1]:
```
from PT3S import Mx
```

In [2]:
```
mx=Mx.Mx(r'..\testdata\M-1-0-1.MX1')
```

with Pandas

MX-Channel TIMESTAMP (1st MX-Channel) used as DataFrame Index; all other MX-Channels are DataFrame Columns

In [3]:
```
import pandas as pd
```

DataFrame

In [4]:
```
df=mx.getPdfFromMxsZipFile(mxsZipFile=r'..\testdata\M-1-0-1.ZIP')
type(df)
```

Out[4]:
```
pandas.core.frame.DataFrame
```

Series

In [5]:
```
ts=df.iloc[1:-2,0]
type(ts)
```

Out[5]:
```
pandas.core.series.Series
```

Timestamps

In [6]:
```
ts.index[0]
```

Out[6]:
```
Timestamp('2017-06-18 23:00:00')
```

In [7]:
```
ts.index[-1]
```

Out[7]:
```
Timestamp('2017-06-19 00:30:00')
```

DatetimeIndex

In [8]:
```
type(ts.index)
```

Out[8]:
```
pandas.tseries.index.DatetimeIndex
```

Time Zone (tz_convert(None) will remove timezone after converting to UTC time)

In [9]:
```
print(ts.index.tz)
```

```
None
```

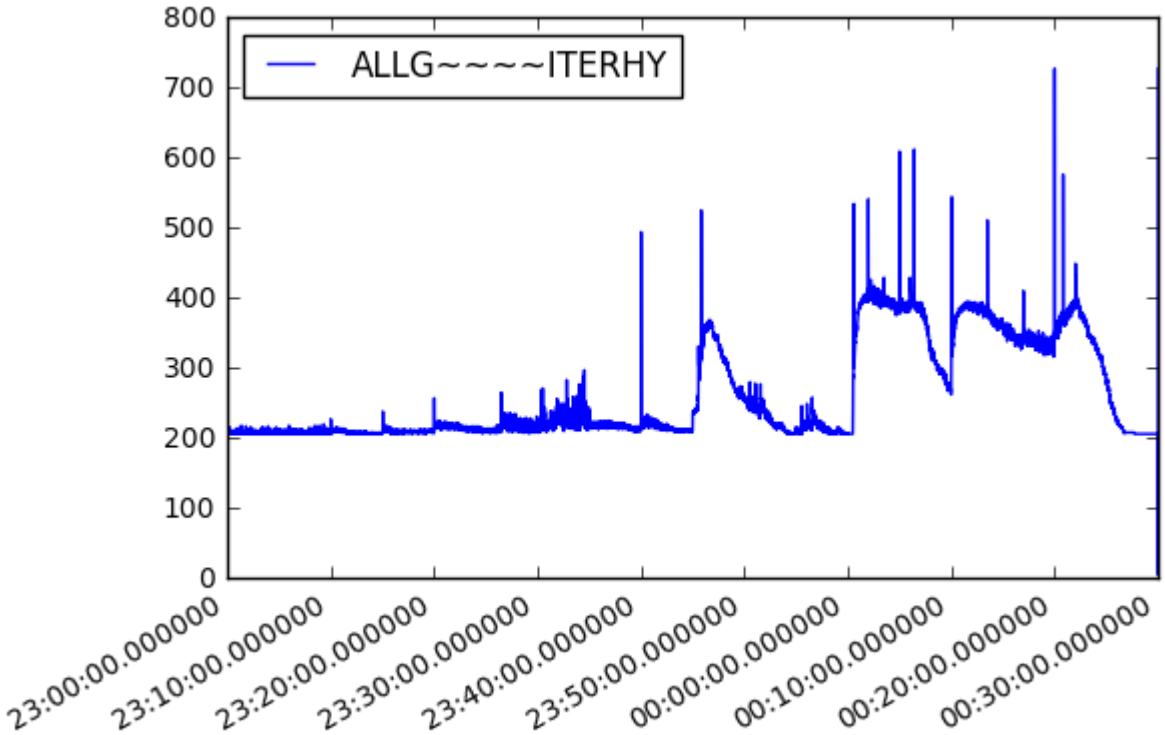
Time Series Plot

In [10]:
```
%matplotlib inline
import matplotlib.pyplot as plt
```

by MX-Channel-Idx

In [11]:
```
plt.figure()
df.iloc[:,17].plot()
plt.legend(loc='best')
```

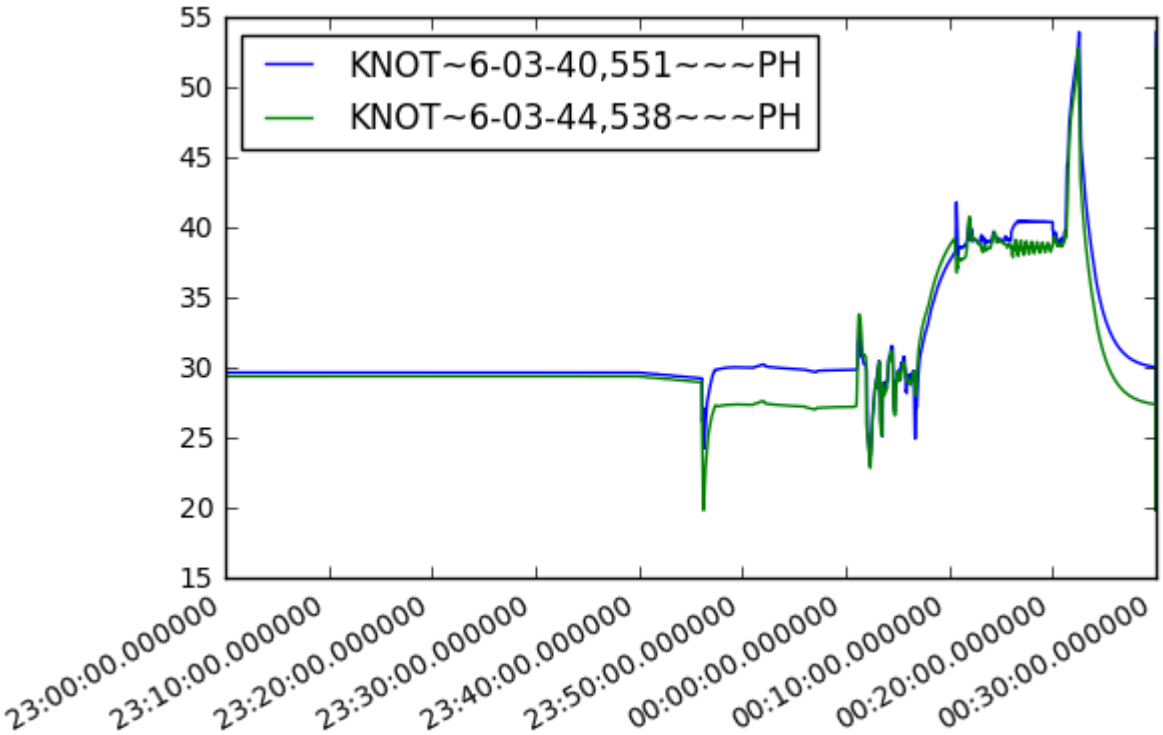
Out[11]: <matplotlib.legend.Legend at 0x1b6a4cf8550>



by MX-Channel-Sir3sID

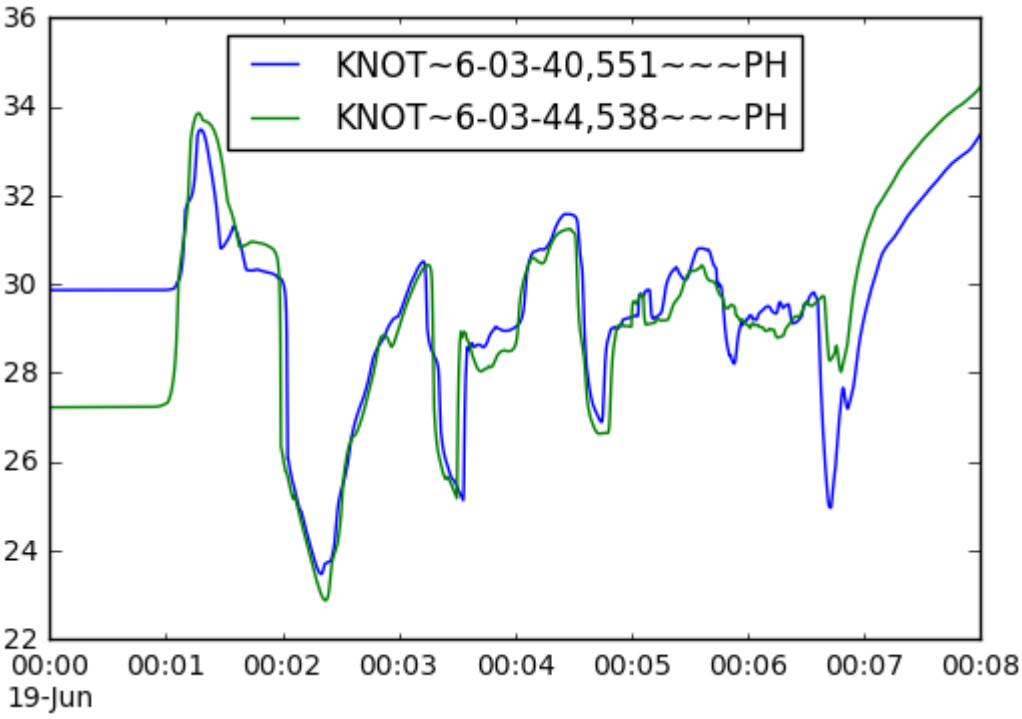
In [13]:
```
import re
regExpCompiled=re.compile(r'^KNOT~6-03-4\d+(\S+)~(\S*)~(\S*)~PH')
df_colsel=df.loc[:,sorted([column for column in df.columns if regExpCompiled.search(c
olumn) != None])]
df_colsel.plot()
```

Out[13]: <matplotlib.axes._subplots.AxesSubplot at 0x1b6b22e8860>



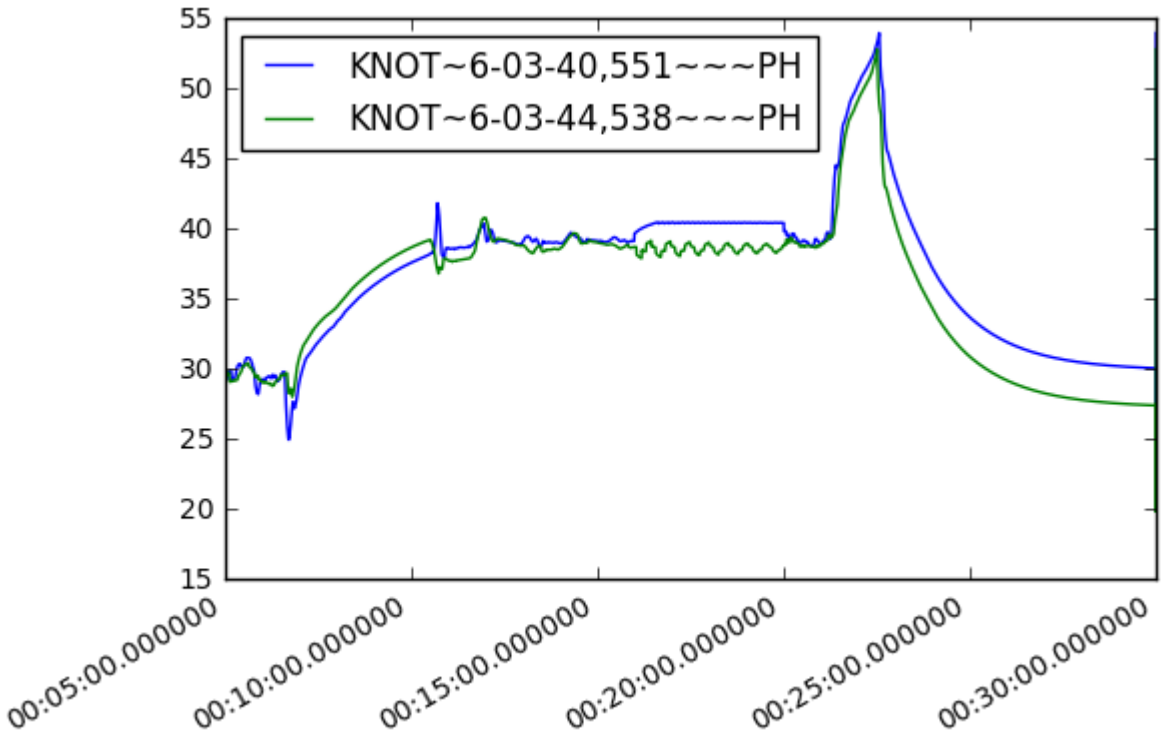
In [14]:
```
start=df_colsel.index[0].to_datetime()+pd.DateOffset(minutes=60)
end=start+pd.DateOffset(minutes=8)
df_colsel.between_time(start_time=start.time(), end_time=end.time()).plot()
```

Out[14]: <matplotlib.axes._subplots.AxesSubplot at 0x1b6c8f90a20>



In [15]:
```
df_colsel[pd.datetime(2017,6,19,0,5):].plot()
```

Out[15]:
```
<matplotlib.axes._subplots.AxesSubplot at 0x1b6bb9f4208>
```



SnapShotType

In [16]:
```
len(df)
```

Out[16]:
```
21604
```

In [17]:
```
df.loc[:,[r'ALLG~~~~SNAPSHOTTYPE']].head(1)
```

Out[17]:

	ALLG~~~~SNAPSHOTTYPE
2017-06-18 23:00:00	b'STAT'

In [18]:
```
df.loc[:,[r'ALLG~~~~SNAPSHOTTYPE']].tail(2)
```

Out[18]:

	ALLG~~~~SNAPSHOTTYPE
2017-06-19 00:30:00	b'TMIN'
2017-06-19 00:30:00	b'TMAX'

In [19]:
```
len(df[df[r'ALLG~~~~SNAPSHOTTYPE'].isin([b'STAT',b'TMIN',b'TMAX'])])
```

Out[19]:
```
3
```

In [20]:
```
df.loc[:,
[r'ALLG~~~~SNAPSHOTTYPE',r'ALLG~~~~ITERHY']].groupby(r'ALLG~~~~SNAPSHOTTYPE').count()
```

Out[20]:

	ALLG~~~~ITERHY
ALLG~~~~SNAPSHOTTYPE	
b'STAT'	1
b'TIME'	21601
b'TMAX'	1
b'TMIN'	1

In [21]:
```
df2=df[df[r'ALLG~~~~SNAPSHOTTYPE'].isin([b'TIME'])]
```

In [22]:
```
len(df2)
```

Out[22]:
```
21601
```

In [23]:
```
df2.loc[:,'KNOT~6-03-44,538~~~PH'].head(4)
```

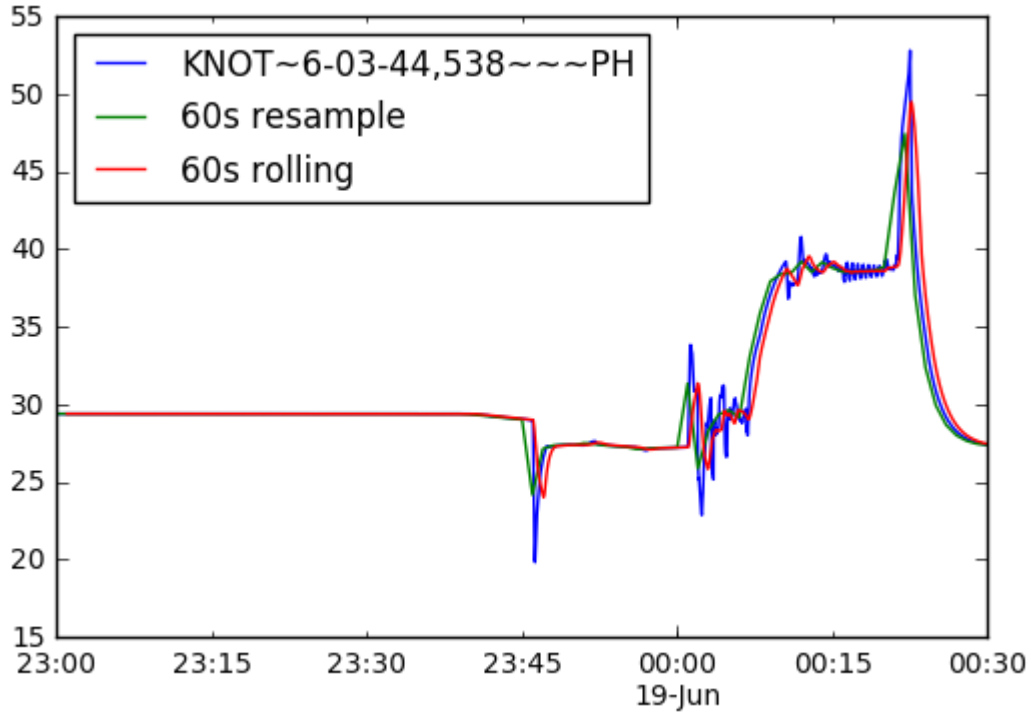
Out[23]:
```
2017-06-18 23:00:00.000    29.404100
2017-06-18 23:00:00.250    29.404099
2017-06-18 23:00:00.500    29.404100
2017-06-18 23:00:00.750    29.404100
Name: KNOT~6-03-44,538~~~PH, dtype: float64
```

Time Series Functions

In [24]:
```
ts=df2.loc[:,'KNOT~6-03-44,538~~~PH']
```

In [25]:
```
ts60=df2.loc[:,'KNOT~6-03-44,538~~~PH'].resample('60s').mean()
ts60.name='60s resample'
ts60r = ts.rolling(window=60*4).mean()
ts60r.name='60s rolling'
plt.figure()
ts.plot()
ts60.plot()
ts60r.plot()
plt.legend(loc='best')
```

Out[25]:
```
<matplotlib.legend.Legend at 0x1b6c231ef60>
```



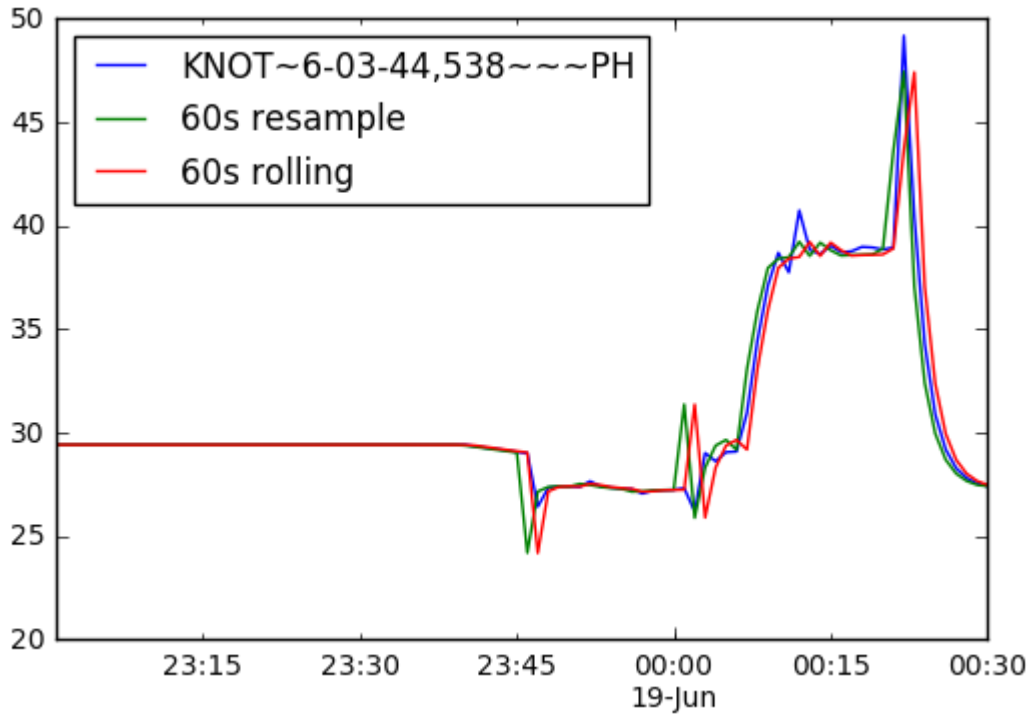
In [26]:
```
df3=pd.DataFrame({ts60.name : ts60,ts.name : ts,ts60r.name : ts60r})
df3=df3.reindex_axis([ts.name,ts60.name,ts60r.name], axis=1)
df3.head()
```

Out[26]:

	KNOT~6-03-44,538~~~PH	60s resample	60s rolling
2017-06-18 23:00:00.000	29.404100	29.404088	NaN
2017-06-18 23:00:00.250	29.404099	NaN	NaN
2017-06-18 23:00:00.500	29.404100	NaN	NaN
2017-06-18 23:00:00.750	29.404100	NaN	NaN
2017-06-18 23:00:01.000	29.404099	NaN	NaN

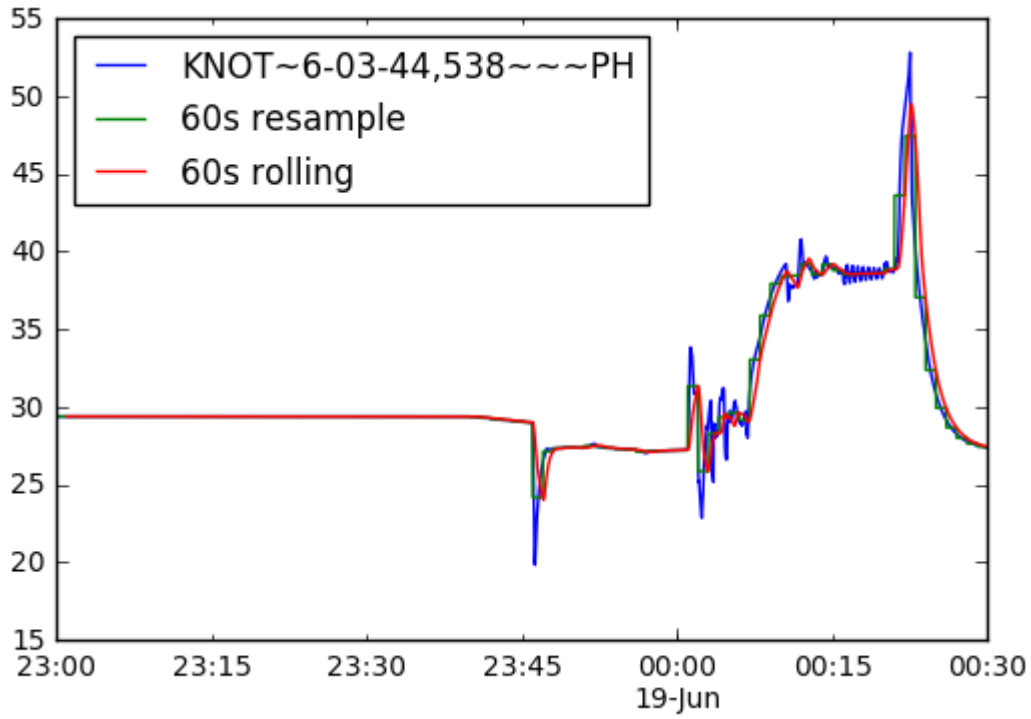
In [27]:
```
df3.dropna().plot()
```

Out[27]: <matplotlib.axes._subplots.AxesSubplot at 0x1b6c89a3320>



In [28]:
```
df3.ffill().plot()
```

Out[28]: <matplotlib.axes._subplots.AxesSubplot at 0x1b6da364f28>



In [ ]: